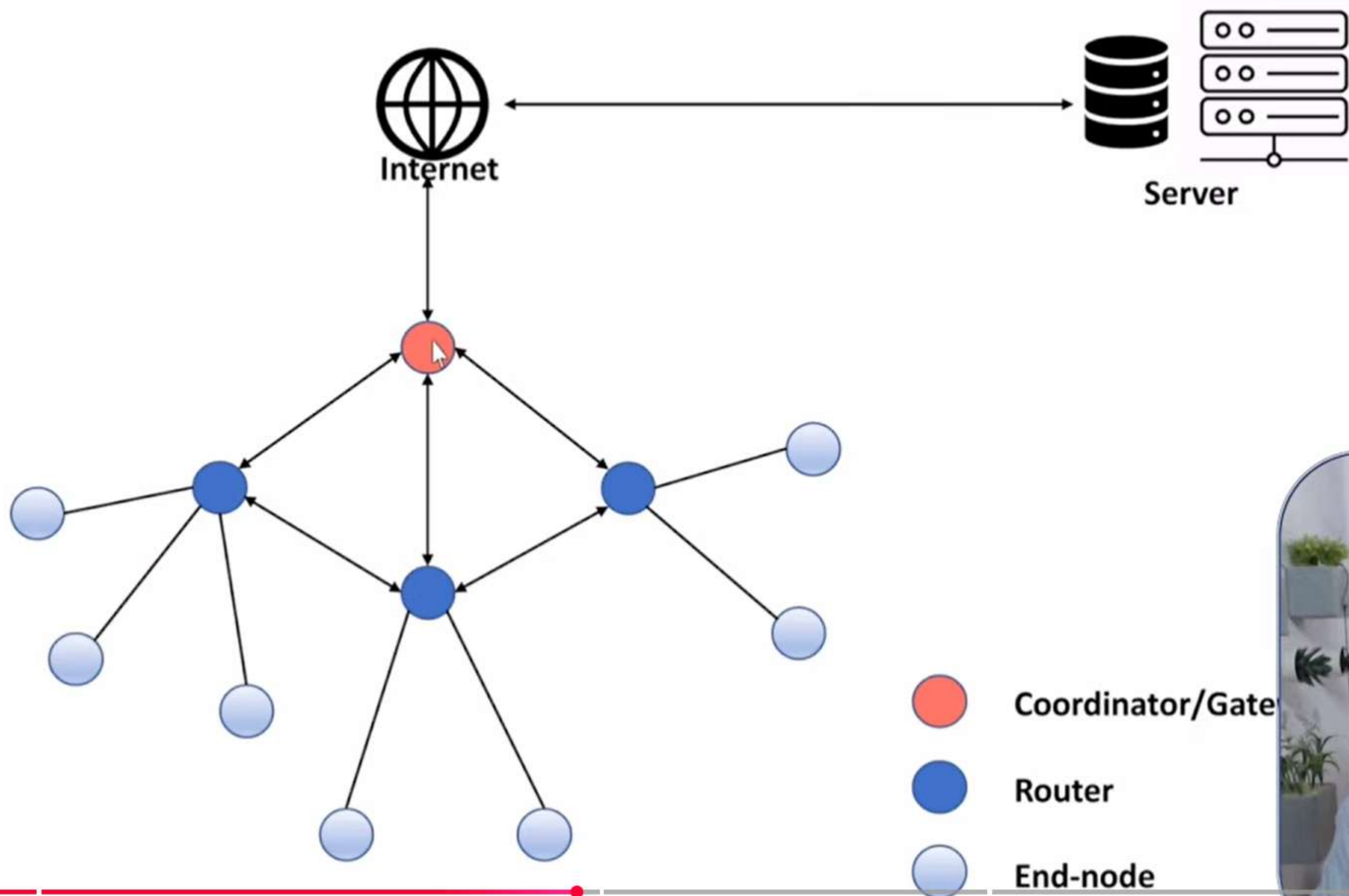


Wireless Sensor Networks





Basics of Wireless Sensor Networks

- A Wireless Sensor Network (WSN) comprises of distributed devices with sensors that are used to monitor the environmental and physical conditions.
- A WSN consists of a number of end nodes and routers and a coordinator.
- End nodes have several sensors attached to them. End nodes can also act as routers.
- Routers are responsible for routing the data packets from end nodes to the coordinator.
- The coordinator collects the data from all the nodes.
- The Coordinator also acts as a gateway that connects the WSN to the Internet.



Characteristics of Wireless Sensor Networks

- Wireless Protocols (ZigBee, Bluetooth, Wi-Fi etc)
- Range (10m to 100m)
- Speed (100KBps to Few MBps)
- Operating Frequency (ISM Bands, 2.4GHz in majority devices)
- Low Power Operating nodes.
- WSNs are self-organizing networks.
- Robust structure. (Even if some nodes are failing, it does autoconf nodes)



Examples of Wireless Sensor Networks

- Weather monitoring systems use WSNs in which the nodes collect temperature, humidity and other data, which is aggregated and analyzed.
- Indoor air quality monitoring systems use WSNs to collect data on the indoor air quality and concentration of various gases.
- Soil moisture monitoring systems use WSNs to monitor soil moisture at various locations.
- Surveillance systems use WSNs for collecting surveillance data (such as motion detection data)
- Smart grids use WSNs for monitoring the grid at various points.



Basics of ZigBee

- ZigBee is developed by ZigBee Alliance, ZigBee is a competitor of 6LOWPAN.
- ZigBee uses IEEE 802.15.4 at MAC and Physical Layer.
- ZigBee is designed for low-cost and low-power wireless IoT Networks.
- ZigBee is used in low data rate applications that requires long battery life and secure networking.
- ZigBee is simpler and less expensive than other WPANs such as Bluetooth and Wi-Fi.
- ZigBee operates at short range around 10m to 20m, networking range can be extended up to 500m.
- ZigBee operates at 2.4GHz unlicensed band.
- ZigBee operates at max data rate of 250 kbps.



Protocol Stack of ZigBee

Application Layer

ZigBee

Network Layer (Routing/ Network)

ZigBee

Data Link Layer (MAC – CSMA/CA)

IEEE 802.15.4

Physical Layer (2.4GHz, 250kbps)

IEEE 802.15.4

▪ ZigBee supports different network topologies like:

- STAR
- Peer to Peer (Mesh)
- Cluster Tree

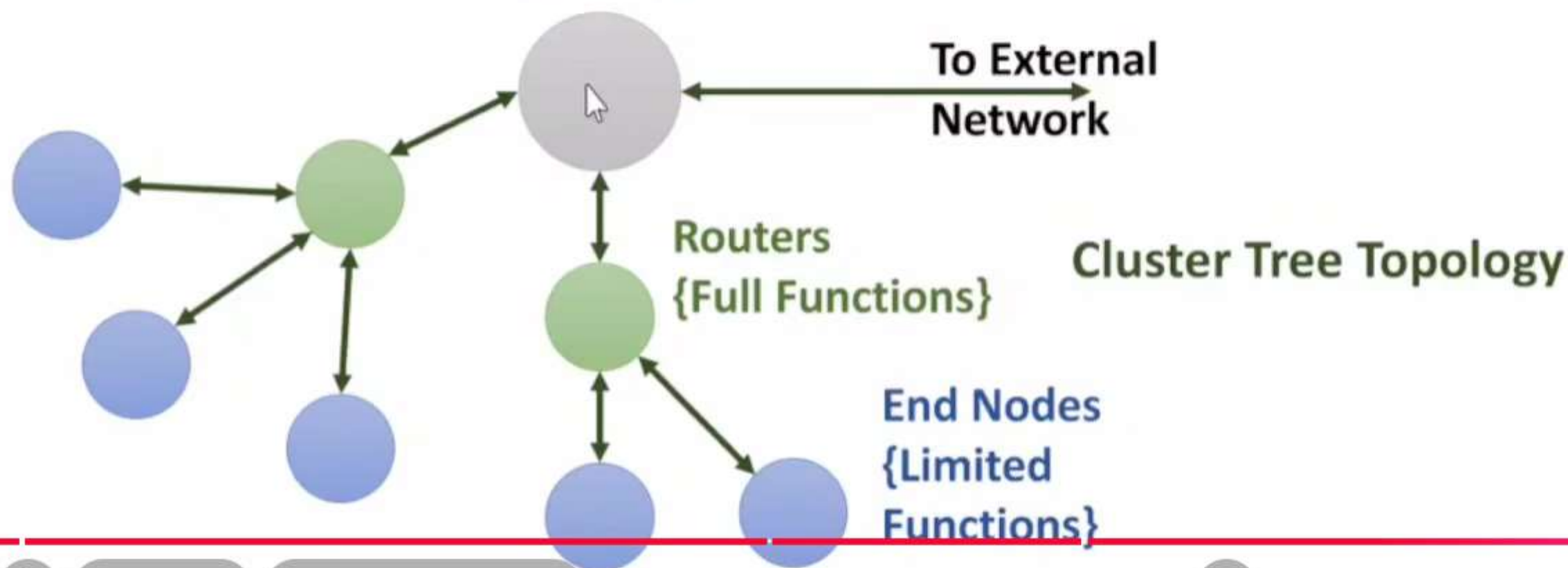
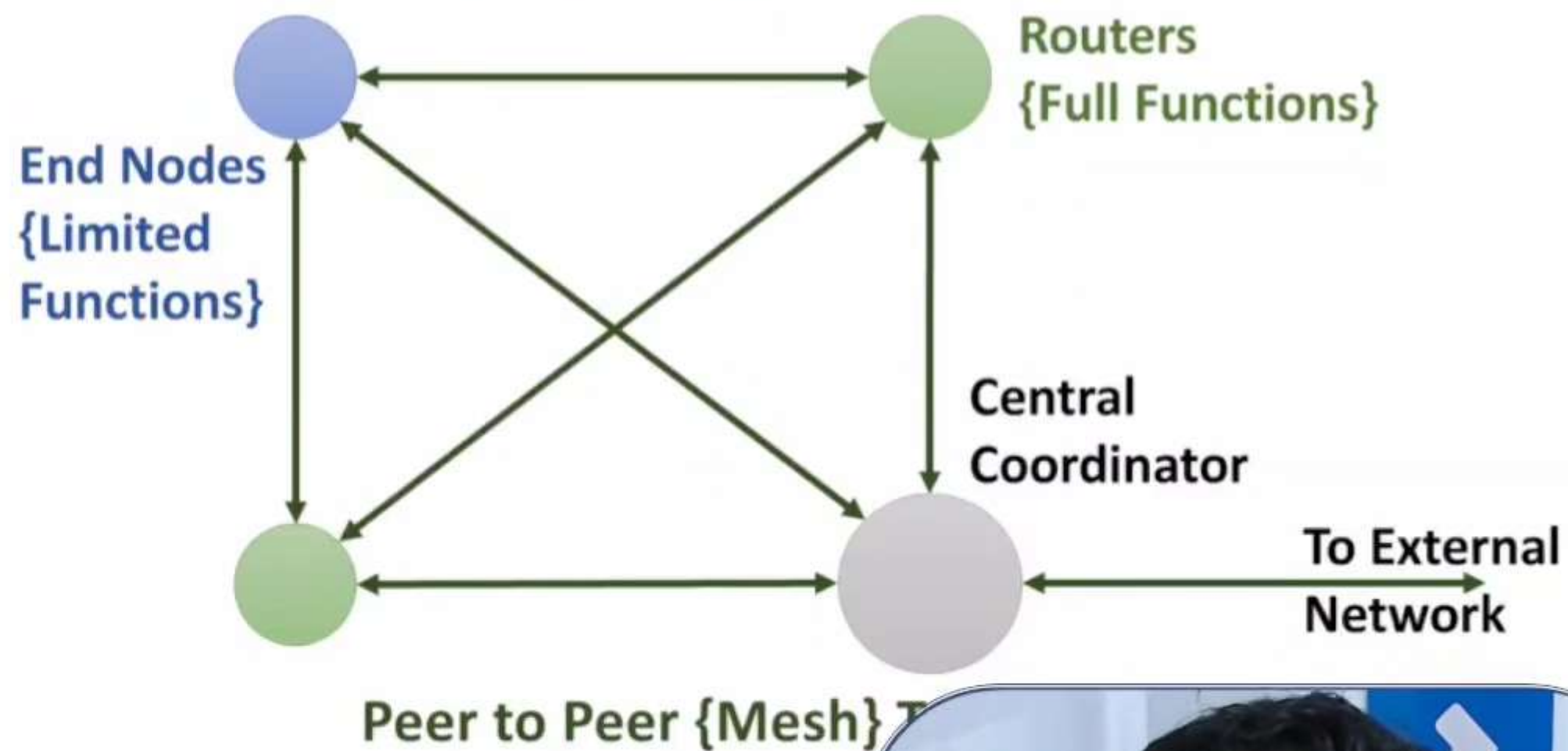
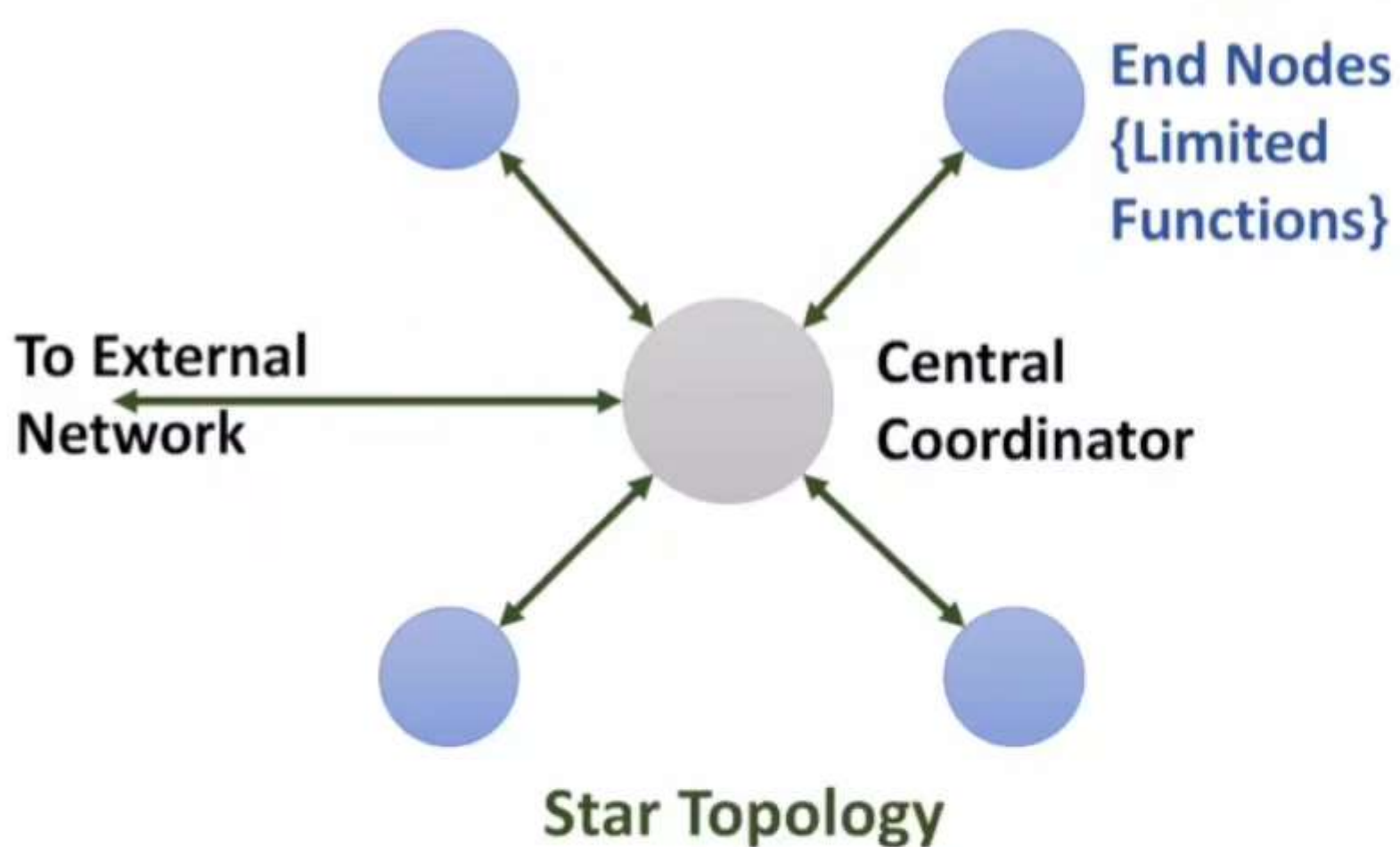
▪ ZigBee can be specified as three different devices:

- ZigBee Coordinator
- ZigBee Router (Full Function Device)

ZigBee End Device (Reduced Function Device)



Topologies used in ZigBee



ZigBee Devices

■ ZigBee Coordinator

- In ZigBee Network, it has one coordinator node, that can store information.
- This Node can control a complete network, that can communicate with an external network.
- ZCs and ZRs require high power, so it can not be battery-powered.

■ ZigBee Router {Full Function Device}

- In ZigBee Network, it can route the data with interconnected nodes.
- It needs less memory than ZCs.
- It can also act as a coordinator in the network.

■ ZigBee End Device {Reduced Function Device}

- In ZigBee Network, it is the least expensive node with minimum
- It has a minimum power requirement, so it can operate with the
- It can only communicate with connected ZR or ZC.
- It communicates periodically for effective battery utilization.



ZigBee Advantages

- Easy to install and implement.
- It has low cost, low power, and long battery life.
- It supports many nodes (around 6500).
- It is more reliable and self-healing.



ZigBee Disadvantages

- It has a low data rate. {250kbps MAX}
- It is not as secure as Wi-Fi and Bluetooth.
- It requires additional devices ZCs and ZRs, which increases the cost.
- It lacks internet protocol support.
- ZigBee Networks are incompatible with other networks.



Explain compare and contrast wired and wireless sensor networks. Explain different topologies for WSN

1. Compare and Contrast: Wired vs. Wireless

Point of Comparison	Wired Sensor Network	Wireless Sensor Network	
Communication Medium	Uses physical cables	Uses wireless signals (RF)	
Installation	Difficult and time-consuming	Easy and fast deployment	
Cost	High installation cost	Lower installation cost	
Reliability	More reliable and stable	Less reliable due to interference	
Mobility	No mobility	Supports mobility	
Scalability	Difficult to expand	Easily scalable	
Power Supply	Continuous power via cables	Battery-powered	
Maintenance	Difficult due to cables	Easier but needs battery replacement	
Interference	Less affected	More affected by noise and obstacles	
Deployment Area	Suitable for fixed environments	Suitable for remote and dynamic areas	

↓

2. Topologies for WSN in IoT

Topology defines how sensor nodes are arranged and how they communicate with the gateway (base station).

Star Topology

In a star network, every sensor node connects directly to a single central gateway or hub.

- **How it Works:** Nodes act as "slaves" and only talk to the "master" hub.
- **Pros:** Simplest to manage; if one node fails, the rest of the network remains unaffected. It consumes the least power for the sensor because it only sends data one hop.
- **Cons:** The central hub is a **single point of failure**; if it goes down, the entire network dies. Range is limited to the signal reach of the hub.

Mesh Topology

Nodes are interconnected, allowing data to "hop" from one sensor to another until it reaches the gateway.

- **How it Works:** Every node can act as a router for other nodes.
- **Pros: High Redundancy.** If one path is blocked or a node fails, the network automatically reroutes data (self-healing). It can cover massive areas by adding more nodes.
- **Cons:** High complexity and "overhead" (nodes must stay awake to route others' data), which drains batteries faster.

Tree (Cluster-Tree) Topology

A hybrid structure where nodes are grouped into clusters, each with a "Cluster Head" (CH).

- **How it Works:** Simple nodes talk to their CH, and CHs talk to each other or the main gateway in a hierarchical fashion.
- **Pros:** Combines the power efficiency of a star (for leaf nodes) with the range of a mesh. Highly scalable for massive IoT deployments (e.g., smart agriculture).
- **Cons:** If a Cluster Head fails, the entire branch (cluster) is disconnected from the network.

Summary of Selection

- **Choose Wired** for: Industrial control systems, security cameras, or when sensors are in environments with thick concrete/metal that blocks signals.
- **Choose Wireless (Star)** for: Home automation or small offices where all devices are close to a router (WiFi/Zigbee).
- **Choose Wireless (Mesh/Tree)** for: Smart cities, agriculture, and large-scale environmental monitoring where coverage needs to span kilometers.

Access Technologies and Their Application Areas

In the ecosystem of the Internet of Things (IoT), selecting the right communication protocol is vital for balancing power consumption, range, and data throughput. Below is a detailed overview of Zigbee and Bluetooth Low Energy (BLE).

1) Zigbee

Explanation:

Zigbee is a low-power, low-data-rate wireless communication technology based on the **IEEE 802.15.4** standard. It is specifically designed for short-range communication and is highly versatile, supporting mesh, star, and tree topologies. Zigbee's primary focus is providing reliable and energy-efficient communication between numerous devices.

Features:

- **Low power consumption:** Ideal for battery-operated devices.
- **Low data rate:** Supports speeds up to 250 kbps.
- **Supports mesh networking:** Allows for self-healing and extended range via intermediate nodes.
- **Short range:** Typically covers 10–100 meters.
- **High scalability:** Can support thousands of nodes in a single network.

Application Areas:

- **Home automation:** Smart lights, smart switches, and security sensors.
- **Industrial monitoring and control:** Process automation and machine health tracking.
- **Smart energy meters:** Remote reading and grid management.
- **Building automation systems:** HVAC control and lighting management.
- **Wireless sensor networks:** Environmental monitoring and data collection.

2) BLE (Bluetooth Low Energy)

Explanation:

BLE is a wireless communication technology introduced as part of the Bluetooth 4.0 specification. Unlike classic Bluetooth, BLE is optimized for very low power consumption, making it perfect for devices that transmit small amounts of data periodically and need to maintain long battery life.

Features:

- **Very low power consumption:** Can run for months or years on a coin-cell battery.
- **Moderate data rate:** Supports speeds up to 1 Mbps or 2 Mbps in newer versions.
- **Short range:** Effective range up to ~100 meters in open space.
- **Fast connection setup:** Connects and sends data much faster than classic Bluetooth.
- **Smartphone compatibility:** Native support on almost all modern mobile devices.

Application Areas:

- **Wearables:** Fitness trackers and smartwatches.
- **Healthcare devices:** Heart rate monitors and glucose sensors.
- **Smart home devices:** Door locks and proximity sensors.
- **Location-based services:** Beacons for indoor navigation or retail marketing.
- **Mobile accessories:** Wireless keyboards and specialized peripherals.

Conclusion

While both technologies focus on efficiency, **Zigbee** is best suited for large-scale, low-power mesh networks required in industrial and whole-home settings. Conversely, **BLE** is ideal for personal, short-range devices that require seamless integration with smartphones.

Write a short note on MEMS.

MEMS (Micro-Electro-Mechanical Systems)

MEMS are very small devices that integrate **mechanical components, sensors, actuators, and electronics** on a single chip. They are fabricated using microfabrication technology and operate at a microscopic scale.

Components of MEMS:

- **Microsensors** – sense physical parameters like temperature, pressure, motion
- **Microactuators** – perform actions based on signals
- **Microelectronics** – process the data

Features:

- Small size and lightweight
- Low power consumption
- High precision and reliability
- Low cost in mass production

Applications:

- Smartphones (accelerometer, gyroscope)
- Automobiles (airbag sensors)
- Medical devices (biosensors)
- Industrial systems (pressure sensors)

Here is your **same content regenerated with proper formatting (no changes in content)** from your document :

Define sensors in IoT. Give classification of sensor and explain any four type.

Explain the different types of sensors are used for measuring one of the physical properties and give its representative examples.

Sensors in IoT

Definition:

A sensor is a device that detects physical or environmental conditions such as temperature, light, motion, pressure, etc., and converts them into electrical signals. In IoT systems, sensors collect data from the real world and send it to other devices for processing and decision-making.

Classification of Sensors (with 1-line explanation)

1. Based on Output

- Analog Sensors: Provide continuous output values (e.g., temperature changing slowly).
- Digital Sensors: Provide discrete output (ON/OFF or 0 and 1).

2. Based on Energy Source

- Active Sensors: Require external power to operate.

- **Passive Sensors:** Work without external power by using energy from the environment.

3. Based on Application

- Sensors are classified based on what they measure, such as temperature, light, motion, pressure, etc.
-

Explanation of Any Four Types of Sensors (Daily Life Examples)

1. Temperature Sensor

A temperature sensor measures the temperature of surroundings and converts it into an electrical signal.

Daily Example: Used in air conditioners and refrigerators to control temperature automatically.

Application: Weather monitoring, cooling systems

2. Pressure Sensor

A pressure sensor measures the pressure of air or liquids.

Daily Example: Used in car tires to indicate if air pressure is low.

Application: Automobiles, water supply systems

3. Motion Sensor

A motion sensor detects movement of people or objects.

Daily Example: Automatic lights that turn ON when someone enters a room or corridor.

Application: Security systems, smart homes

4. Light Sensor

A light sensor measures the intensity of light.

Daily Example: Smartphone brightness automatically adjusts based on surrounding light.

Application: Street lighting, mobile devices

5. Humidity Sensors

These sensors measure moisture or humidity in the air.
Examples: Weather apps, room humidity controller
Use: Used in agriculture, weather stations, and homes

Difference (8 Points)			
Point	Sensor	Actuator	
1. Function	Measures physical quantity	Performs action	
2. Role	Input device	Output device	
3. Conversion	Physical → Electrical signal	Electrical → Physical action	
4. Purpose	Data collection	Control and operation	
5. Energy Flow	Receives energy from environment	Uses electrical energy to act	
6. Example Devices	Temperature sensor, LDR	Motor, LED, relay	
7. Use in System	Sends data to controller	Receives command from controller	
8. Output Type	Signal/data	Mechanical movement or action	

What is the smart object? Give the security and privacy concerns of smart object in IoT.

Smart Object in IoT

Definition:

A smart object is a physical device embedded with sensors, processors, communication modules, and software that enables it to sense the environment, process data, and communicate with other devices over a network. Smart objects are the basic building blocks of IoT systems.

Security and Privacy Concerns of Smart Objects in IoT

1. Data Privacy Issues

Smart objects collect personal and sensitive data (like location, health, usage patterns). If not protected, this data can be misused.

2. Unauthorized Access

Weak authentication can allow attackers to access and control smart devices without permission.

3. Data Theft (Eavesdropping)

Data transmitted over networks can be intercepted if proper encryption is not used.

4. Data Tampering (Integrity Issues)

Attackers can modify the data being transmitted, leading to incorrect actions or decisions.

5. Device Hijacking

Hackers can take control of smart objects and use them for malicious activities like spying or botnet attacks.

6. Denial of Service (DoS) Attacks

Attackers can overload devices or networks, making them unavailable for normal use.

7. Insecure Software/Firmware

Outdated or poorly designed software can have vulnerabilities that attackers can exploit.

8. Lack of Standard Security Measures

Different manufacturers follow different security practices, leading to inconsistent protection.

9. Physical Security Risks

Devices placed in open areas can be physically accessed, damaged, or tampered with.

10. Identity Spoofing

Attackers can pretend to be a genuine device and gain access to the system.

Explain characteristics of smart object

Give the trend in smart object.

Characteristics of Smart Objects

A smart object is an intelligent device in IoT that can sense, process, and communicate data. It has the following characteristics:

1. Unique Identity

Each smart object has a unique ID (like IP address or MAC address) which helps in identification in a network.

2. Sensing Capability

It can sense physical parameters such as temperature, light, motion, etc., using sensors.

3. Processing Capability

It has a microcontroller or processor to process the collected data.

4. Communication Capability

It can communicate with other devices using technologies like Wi-Fi, Bluetooth, Zigbee, etc.

5. Actuation Capability

It can perform actions based on commands, such as switching ON/OFF devices.

6. Energy Efficiency

Smart objects consume less power and often use battery or energy-saving techniques.

7. Intelligence

They can make decisions automatically using predefined logic or AI.

Trends in Smart Objects

1. Miniaturization

Smart objects are becoming smaller, compact, and easy to carry.

2. Increased Intelligence

Integration of Artificial Intelligence and Machine Learning is making devices smarter and more autonomous.

3. Energy Harvesting

Devices are using renewable energy sources like solar or vibration, reducing battery dependency.

4. Improved Connectivity

Advanced technologies like 5G, LoRa, and NB-IoT provide faster and more reliable communication.

5. Edge Computing

Data processing is done near the device, reducing delay and improving performance.

6. Enhanced Security

Improved encryption and authentication techniques are being used to protect data.

7. Interoperability and Standardization

Devices from different manufacturers are becoming compatible through common standards.

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